Project 3: Link Analysis

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Abstract

Scientific collaboration networks are an important source of information for understanding the structure of research communities. In these networks, nodes represent authors and edges represent collaborations between them. Studying these networks makes it possible to identify influential researchers and understand patterns of scientific collaboration.

In this project, a large collaboration network derived from the arXiv dataset is analyzed. The goal is to construct a co-authorship graph and apply link analysis techniques to identify the most influential authors in the network.

To reach this objective, a graph is built where nodes correspond to authors and edges represent collaborations between authors who have written at least one paper together. The PageRank algorithm is then applied to measure the importance of each author in the network. Then, results were compared with other simple metrics such as weighted degree, unweighted degree, and number of publications.

1 Introduction

Scientific collaboration networks provide useful insights into the structure of research communities. In these networks, authors are connected when they co-author a paper. Studying this networks makes possible to identify influential researchers and better understand collaboration patterns.

In this project, a co-authorship network is constructed using the arXiv dataset, and the PageRank algorithm is applied to rank authors according to their importance within the network.

2 Dataset

The dataset used in this project is the *arXiv dataset* available on Kaggle. It contains information about scientific papers published on arXiv, including the list of authors for each paper.

Due to the large size of the dataset (more than 5 GB of data and around 1.7 million records), the analysis was done on a subset of 200,000 records. This choice makes the computation more manageable and requires less memory usage, while preserving the validity of the adopted methods for large-scale data.

Each record in the dataset represents a paper and includes fields such as:

- Title
- Abstract
- Authors
- Categories

For this project, only the list of authors was used to construct the co-authorship network.

3 Data Organization

The first step of the project consists of transforming the dataset into a structure suitable for network analysis.

Each paper contains a list of authors. From this information, a co-authorship graph is constructed using the following representation:

- each node corresponds to an author;
- an edge between two nodes indicates that the corresponding authors have collaborated on at least one paper;
- the weight of an edge represents the number of papers written together by the two authors.

For each paper containing authors $a_1, a_2, \ldots, a_k$, edges are created between every pair of authors in the list. This produces a collaboration network in which the structure of scientific interactions can be analyzed.

Since the number of authors and collaborations is very large, the network is stored using a sparse matrix representation. This choice is particularly effective because most pairs of authors do not collaborate, so the adjacency matrix contains a lot of zero values. As a result, memory usage is reduced significantly and graph computations remain efficient even on large datasets.

4 Data Pre-processing

Before constructing the network, several preprocessing steps are applied to clean and standardize the data.

First, author names are normalized in order to reduce inconsistencies caused by different formatting conventions.

Second, duplicate collaborations within the same paper are avoided so that each pair of authors is considered only once per paper.

Another important preprocessing step consists of counting the number of collaborations between authors. Each time two authors appear together in a paper, the weight of the corresponding edge

in the graph is increased. This allows the network to capture not only whether two authors collaborated, but also how frequently they worked together.

Finally, the resulting list of collaborations is converted into a sparse adjacency matrix representing the structure of the co-authorship network.

5 Algorithms

The main algorithm used in this project is PageRank, which is an important link analysis algorithm originally developed for ranking web pages.

The basic idea is that a node is considered important if it is connected to other important nodes. In the context of this project:

- nodes represent authors;
- edges represent collaborations between authors.

An author receives a high PageRank score if they collaborate with other authors who are themselves influential in the network.

The PageRank algorithm is implemented using an iterative approach based on matrix operations. At each iteration, the score vector is updated until the values stabilize. In this project, the convergence was reached after 87 iterations.

In addition to PageRank, other metrics are computed for comparison:

- unweighted degree, which counts the number of distinct collaborators of an author;
- weighted degree, which measures the total number of collaborations;
- paper count, which measures the number of papers written by each author.

These metrics provide useful reference points for interpreting the PageRank results.

From a scalability point of view, the adopted approach is suitable for large datasets because it relies on sparse matrix representations and efficient matrix-vector operations. This means that even if in this case only a subset of the dataset was analyzed, the chosen methods remain appropriate for the full dataset.

6 Experiments

Several experiments were conducted in order to evaluate the behavior of the proposed approach.

First, the collaboration network was constructed from the dataset and the PageRank algorithm was applied to compute an importance score for each author.

After computing PageRank, authors were ranked according to their score. The top ranked authors were then analyzed to better understand the structure of the collaboration network.

In addition, PageRank scores were compared with other metrics like weighted degree, unweighted degree, and paper count. This comparison helps determine whether PageRank captures aspects of the network structure that are not reflected by simpler metrics.

Finally, PageRank was also computed on the binary version of the graph, where all collaboration edges have equal weight. This makes it possible to evaluate the effect of repeated collaborations on the final ranking.

7 Results and Discussion

The results show that PageRank highlights authors occupying structurally central positions in the collaboration network.

Compared to simpler metrics such as degree, PageRank provides a more informative measure of importance because it takes into account not only the number of collaborations, but also the influence of the collaborators.

After computing the additional metrics, differences emerged across the rankings. The top authors according to PageRank do not necessarily coincide with the top authors according to weighted degree or unweighted degree. In some cases, PageRank and paper count produce similar rankings, suggesting that a large number of publications can also influence the PageRank score.

To better examine these differences, the results of the various metrics were merged into a single comparison. Table 1 reports the top 10 authors ranked by PageRank together with their main metric values, while Table 2 shows their positions according to the other ranking measures.

Rank	Author	PageRank	W. degree	UnW. degree	Paper count
1	p. h. vincent	0.000134	343	129	140
2	s. donald p.	0.000089	1390	706	68
3	gehrels n.	0.000081	3048	1294	118
4	canfield p. c.	0.000079	887	310	101
5	n. franco	0.000077	296	122	96
6	g. guang-can	0.000076	295	117	69
7	l. loet	0.000075	99	57	78
8	a. uwe	0.000073	206	66	107
9	c. d. irccyn	0.000070	266	67	116
10	z. tao	0.000069	193	93	58

Table 1: Top 10 authors ranked by PageRank and their main network metrics.

Rank	Author	Rank PR	Rank WD	Rank UnW. D.	Rank PC
1	p. h. vincent	1	1980	1097	3
2	s. donald p.	2	1199	539	25
3	gehrels n.	3	459	309	4
4	canfield p. c.	4	1476	916	8
5	n. franco	5	2027	1104	10
6	g. guang-can	6	2028	1109	24
7	l. loet	7	2224	1169	17
8	a. uwe	8	2117	1160	7
9	c. d. irccyn	9	2057	1159	6
10	z. tao	10	2130	1133	35

Table 2: Ranking positions of the top authors according to different metrics.

The tables show that the authors with the highest PageRank are not necessarily those with the largest number of collaborations or publications. For example, p. h. vincent ranks first according to PageRank despite having relatively low positions in the weighted and unweighted degree rankings. This suggests that PageRank captures the structural importance of an author in the collaboration network by looking not only the number of collaborations, but also the importance of the collaborators. Therefore, PageRank provides a more informative measure of centrality than simple counts of collaborations or publications.

A correlation matrix was also computed in order to compare the different measures. Figure 1 shows the Spearman correlation matrix between PageRank, weighted degree, unweighted degree, and paper count.

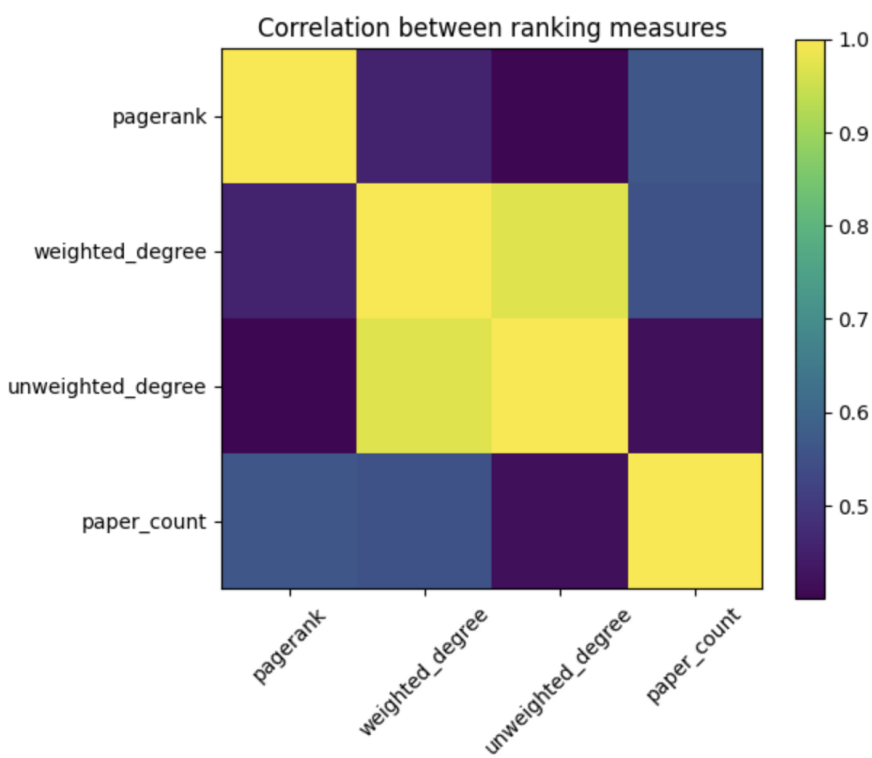

Figure 1: Spearman correlation matrix

As expected, weighted degree and unweighted degree are highly correlated, since both measure the number of collaborations of an author. PageRank shows a moderate correlation with the degree-based measures, indicating that it is influenced by the number of connections but also captures additional structural information about the network. Finally, paper count has a weaker correlation with the other metrics, suggesting that the number of publications alone does not necessarily reflect an author's centrality in the collaboration network.

To evaluate the effect of edge weights, PageRank was also computed on the binary version of the graph, where all collaborations have equal weight. The comparison between weighted and unweighted PageRank shows a high Spearman correlation, indicating that the overall ranking is largely consistent, although small differences may appear due to the influence of repeated collaborations.

8 Conclusion

In this project, a co-authorship network was constructed from the arXiv dataset and analyzed using several ranking measures, including PageRank, weighted degree, unweighted degree, and paper count.

The results show that simple metrics such as the number of publications or collaborations do not fully capture the structural importance of authors in the network.

In contrast, PageRank provides a more informative measure of influence by taking into account not only the number of connections, but also the importance of the collaborators. The comparison between different metrics shows that these measures can reveal structural properties of scientific collaboration that are not visible when considering only productivity indicators.

Overall, the analysis demonstrates the usefulness of link analysis techniques for studying large-scale collaboration networks.

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